Probabilistic Combinatorics

Sheet 9

June 17, 2024

Problem 1. Let L be a finite distributive lattice, $\mu_1, \mu_2 : L \to \mathbb{R}$ functions satisfying

$$\mu_1(x)\mu_2(x) \geq 0 \quad \text{for } x \in L$$
$$\mu_1(x)\mu_2(y) \leq \mu_1(x \vee y)\mu_2(x \wedge y) \quad \text{for } x, y \in L$$

and

$$\sum_{x \in L} \mu_1(x) = \sum_{x \in L} \mu_2(x).$$

Show that for any monotonically increasing function $f : L \to \mathbb{R}$ it holds that

$$\sum_{x \in L} f(x)\mu_1(x) \geq \sum_{x \in L} f(x)\mu_2(x).$$

Proof.
Let $a = -\min\{f(x) | x \in L\}$. Define the new function $f'(x) = f(x) + a$, it is obvious that $f'(x) \geq 0$ for all $x \in L$. Next, we define the following four functions

$$\alpha = \mu_1 \qquad \beta = f' \cdot \mu_2 \qquad \gamma = f' \cdot \mu_1 \qquad \delta = \mu_2$$

Based on this definition, for $x, y \in L$, we have

$$\alpha(x)\beta(y) = \mu_1(x)f'(y)\mu_2(y)$$

Since $\mu_1(x)\mu_2(y) \leq \mu_1(x \vee y)\mu_2(x \wedge y)$ for $x, y \in L$, then the equation is equivalent to

$$\begin{aligned}\alpha(x)\beta(y) &\leq f'(y)\mu_1(x \vee y)\mu_2(x \wedge y) \\ &\leq f'(x \vee y)\mu_1(x \vee y)\mu_2(x \wedge y) \\ &= \gamma(x \vee y)\delta(x \wedge y)\end{aligned}$$

Therefore, the four functions defined above satisfy the Four Functions Theorem, and thus $\alpha(A)\beta(B) \leq \gamma(A \vee B)\delta(A \wedge B)$ for any $A, B \subseteq L$. Hence, for $A = B = L$, we have

$$\alpha(L)\beta(L) \leq \gamma(L)\delta(L) \tag{1.1}$$

Since $\alpha(L) = \delta(L)$, as $\sum_{x \in L} \mu_1(x) = \sum_{x \in L} \mu_2(x)$, then the inequality (1.1) implies that

$$\begin{aligned}& \beta(L) \leq \gamma(L) \\ \Leftrightarrow \quad & \sum_{x \in L} f'(x)\mu_2(x) \leq \sum_{x \in L} f'(x)\mu_1(x) \\ \Leftrightarrow \quad & \sum_{x \in L} (f(x) + a)\mu_2(x) \leq \sum_{x \in L} (f(x) + a)\mu_1(x) \\ \Leftrightarrow \quad & \sum_{x \in L} f(x)\mu_2(x) + a\sum_{x \in L} \mu_2(x) \leq \sum_{x \in L} f(x)\mu_1(x) + a\sum_{x \in L} \mu_1(x) \qquad (1.2)\end{aligned}$$

Since $\sum_{x\in L}\mu_2(x) = \sum_{x\in L}\mu_1(x)$, then the inequality (1.2) is equivalent to

$$\sum_{x\in L} f(x)\mu_1(x) \geq \sum_{x\in L} f(x)\mu_2(x).$$

, and the proof is finished. □

Problem 2. Let $L_{n,m} = [n]^m$ be the set of all m-tuples of elements of $[n]$. Let

$$(s_1, \ldots, s_m) \leq (s_1^{'}, \ldots, s_m^{'})$$

for $(s_1, \ldots, s_m), (s_1^{'}, \ldots, s_m^{'}) \in [n]^m$, iff $s_i \leq s_i^{'}$ for all i.

Show:

(i) $L_{n,m}$ is a distributive lattice. Describe the $\vee$ and $\wedge$ operations in $L_{n,m}$.

(ii) The subset $S_{n,m} \subseteq L_{n,m}$ of all $(s_1, \ldots, s_m)$ with $s_1 < \ldots < s_m$ equipped with the same order is also a distributive lattice.

(iii) Let now $n \geq m$. There is a bijection $\phi : S_{n,m} \to S_{n,n-m}$, s.t. for all $x, y \in S_{n,m}$, $\phi(y) \leq \phi(x)$ if and only if $x \leq y$.

Proof.
(i): In order to show that $L_{n,m}$ is a distributive lattice, we want to show that $L_{n,m}$ satisfies the following properties:

(a) The binary relation $(L_{n,m}, \leq)$ is a partial order on $L_{n,m}$. That is, we are going to verify the following conditions

- (Reflexivity) $(s_1, \ldots, s_m) \leq (s_1, \ldots, s_m)$ for all $(s_1, \ldots, s_m) \in L_{n,m}$. This is true since $s_i \leq s_i$ for $1 \leq i \leq m$.
- (Antisymmetry) $(s_1, \ldots, s_m) \leq (s_1^{'}, \ldots, s_m^{'})$ and $(s_1^{'}, \ldots, s_m^{'}) \leq (s_1, \ldots, s_m)$ for $(s_1, \ldots, s_m), (s_1^{'}, \ldots, s_m^{'}) \in L_{n,m}$ implies $(s_1, \ldots, s_m) = (s_1^{'}, \ldots, s_m^{'})$. This is true since $s_i \leq s_i^{'}$ and $s_i^{'} \leq s_i$ for $1 \leq i \leq m$ implies $s_i = s_i^{'}$.
- (Transitivity) $(s_1, \ldots, s_m) \leq (s_1^{'}, \ldots, s_m^{'})$ and $(s_1^{'}, \ldots, s_m^{'}) \leq (s_1^{''}, \ldots, s_m^{''})$ for $(s_1, \ldots, s_m), (s_1^{'}, \ldots, s_m^{'}), (s_1^{''}, \ldots, s_m^{''}) \in L_{n,m}$ implies $(s_1, \ldots, s_m) \leq (s_1^{''}, \ldots, s_m^{''})$. This is true since $s_i \leq s_i^{''}$ when $s_i \leq s_i^{'}$ and $s_i^{'} \leq s_i^{''}$ for $1 \leq i \leq m$.

Therefore, the binary relation $(L_{n,m}, \leq)$ is a partial order on $L_{n,m}$.

(b) The partial order $(L_{n,m}, \leq)$ is a lattice. Let $p = (s_1, \ldots, s_m) \in L_{n,m}$ and $q = (s_1^{'}, \ldots, s_m^{'}) \in L_{n,m}$, we want to verify the following conditions

- The largest element $r = (s_1^{''}, \ldots, s_m^{''}) \in L_{n,m}$ such that $r \leq p, q$ is unique. It is obvious that r is the largest element in $L_{n,m}$ such that $r \leq p, q$ when $s_i^{''} = \min(s_i, s_i^{'})$, and such r is unique. That is, $p \wedge q = r$, and

$$p \wedge q = \left(\min(s_i, s_i^{'})\right)_{i=1,\ldots,m}$$

- The smallest element $r = (s_1^{''}, \ldots, s_m^{''}) \in L_{n,m}$ such that $r \geq p, q$ is unique. It is obvious that r is the smallest element in $L_{n,m}$ such that $r \geq p, q$ when $s_i^{''} = \max(s_i, s_i^{'})$, and such r is unique. That is, $p \vee q = r$, and

$$p \vee q = \left(\max(s_i, s_i^{'})\right)_{i=1,\ldots,m}$$

Therefore, the partial order $(L_{n,m}, \leq)$ is a lattice.

(c) The lattice $L_{n,m}$ is distributive. Let $p = (s_1, \ldots, s_m) \in L_{n,m}$, $q = (s_1^{'}, \ldots, s_m^{'}) \in L_{n,m}$, and $r = (s_1^{''}, \ldots, s_m^{''}) \in L_{n,m}$, we want to verify the following conditions

- $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$. That is, we want to prove the following equation holds

$$\max(s_i, \min(s_i^{'}, s_i^{''})) = \min(\max(s_i, s_i^{'}), \max(s_i, s_i^{''})) \tag{2.1}$$

  In case that $s_i \leq \min(s_i^{'}, s_i^{''})$, then the LHS and RHS of the equation (2.1) are equal to $\min(s_i^{'}, s_i^{''})$, and thus the equation holds in this case.

  Otherwise, when $s_i \geq \min(s_i^{'}, s_i^{''})$, then the LHS of the equation (2.1) is equal to s_i, and s_i is larger than either s_i or $s_i^{'}$, or s_i is larger than both. If s_i is larger than either $s_i^{'}$ or $s_i^{''}$, then the RHS of the equation (2.1) is equal to either $\min(s_i, s_i^{''})$ or $\min(s_i^{'}, s_i)$ respectively, and thus the RHS of the equation is equal to either $\min(s_i, s_i^{''}) = s_i$ or $\min(s_i^{'}, s_i) = s_i$; hence, the LHS and RHS of the equation (2.1) are equal to each other in this case. If s_i is larger than both $s_i^{'}$ and $s_i^{''}$, then the RHS of the equation is equal to $\min(s_i, s_i) = s_i$, and thus the LHS and RHS of the equation (2.1) are equal to each other in this case. Therefore, the equation (2.1) always holds.

- $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$. This can be shown to be true by using the similar logic described above.

Therefore, the lattice $L_{n,m}$ is distributive, and the proof for (i) is finished.

(ii): Based on the idea of showing that the binary relation $(L_{n,m}, \leq)$ is a partial order on $L_{n,m}$, which is presented in the previous part, it is easy to verify that $(S_{n,m}, \leq)$ is a partial order on $S_{n,m}$. That is, it remains to show that

(a) The partial order $(S_{n,m}, \leq)$ is a lattice. Let $p = (s_1, \ldots, s_m) \in S_{n,m}$ and $q = (s_1^{'}, \ldots, s_m^{'}) \in S_{n,m}$, we want to verify the following conditions

- The largest element $r = (s_1^{''}, \ldots, s_m^{''}) \in S_{n,m}$ such that $r \leq p, q$ is unique. It is obvious that r is the largest element in $L_{n,m}$ such that $r \leq p, q$ when $s_i^{''} = \min(s_i, s_i^{'})$ for $1 \leq i \leq m$, and such r is unique. That is, it remains to determine whether $r \in S_{n,m}$. To be more specific, we want to check whether $s_j^{''} < s_i^{''}$ for $1 \leq j < i \leq m$. Recall that, we are considering $s_j^{''} = \min(s_j, s_j^{'})$ and $s_i^{''} = \min(s_i, s_i^{'})$. It is worth to note that $s_j < s_i$ and $s_j^{'} < s_i^{'}$ for $1 \leq j < i \leq m$, since $p, q \in S_{n,m}$. Next, we consider the following cases
  - If $s_i \leq s_i^{'}$ and $s_j \leq s_j^{'}$, then $s_i^{''} = s_i$ and $s_j^{''} = s_j$. Since $s_j < s_i$, then $s_j^{''} < s_i^{''}$.
  - If $s_i \geq s_i^{'}$ and $s_j \geq s_j^{'}$, then $s_i^{''} = s_i^{'}$ and $s_j^{''} = s_j^{'}$. Since $s_j^{'} < s_i^{'}$, then $s_j^{''} < s_i^{''}$.
  - If $s_i \leq s_i^{'}$ and $s_j \geq s_j^{'}$, then $s_i^{''} = s_i$ and $s_j^{''} = s_j^{'}$. Since $s_j^{'} \leq s_j < s_i$, then $s_j^{''} < s_i^{''}$.
  - If $s_i \geq s_i^{'}$ and $s_j \leq s_j^{'}$, then $s_i^{''} = s_i^{'}$ and $s_j^{''} = s_j$. Since $s_j \leq s_j^{'} < s_i^{'}$, then $s_j^{''} < s_i^{''}$.

  Therefore, $r \in S_{n,m}$, as the inequality $s_j^{''} < s_i^{''}$ always holds for $1 \leq j < i \leq m$. That is, $p \wedge q = r$, and

$$p \wedge q = \left(\min(s_i, s_i^{'})\right)_{i=1,\ldots,m}$$

- The smallest element $r = (s_1^{''}, \ldots, s_m^{''}) \in S_{n,m}$ such that $r \geq p, q$ is unique. It is obvious that r is the smallest element in $L_{n,m}$ such that $r \geq p, q$ when $s_i^{''} = \max(s_i, s_i^{'})$ for $1 \leq i \leq m$, and such r is unique. By similarly applying the logic presented previously, we can conclude that $r \in S_{n,m}$. That is, $p \vee q = r$, and

$$p \vee q = \left(\max(s_i, s_i^{'})\right)_{i=1,\ldots,m}$$

Therefore, the partial order $(S_{n,m}, \leq)$ is a lattice.

(b) The lattice $S_{n,m}$ is distributive. With $p \wedge q$, and $p \vee q$ defined previously for $p, q \in S_{n,m}$, we can show the lattice $S_{n,m}$ is distributive similarly to the way that we show the lattice $L_{n,m}$ is distributive in the statement (i).

Therefore, the lattice $S_{n,m}$ is a distributive lattice. □

Problem 3. Consider the following experiment: For $1 \leq m \leq n$ there are m balls thrown one by one into n baskets. In which baskets the balls land are independent and the j-th ball will land with probability $p_{a_i,j}$ in the basket a_i.

(i) Define the probability space and identify the outcomes with $L_{n,m}$ from (2).

(ii) For $I \subseteq [m]$ and $t_I \in R$ define the event

$$A_I := \{(a_1, \ldots, a_m) \in L_{n,m} \mid \sum_{i \in I} a_i \geq t_I\}.$$

Show that, for $I, J \subseteq [m]$ with $I \cap J = \emptyset$ or $I \cup J = [m]$ and real numbers t_I and t_J it holds that

$$\mathbb{P}(A_I \cap A_J) \geq \mathbb{P}(A_I)\mathbb{P}(A_J).$$

Proof.
(i): The probability space for the experiment is the discrete probability space $(\Omega, \mathbb{P})$ where $\Omega = L_{n,m}$, or the set of all m-tuples of $[n]$ in form of $(a_1, \ldots, a_m)$, where a_i is the basket in which the i-th ball lands. Moreover, the probability measure $\mathbb{P}$ of this discrete probability space is

$$\mathbb{P}\Big((a_1, \ldots, a_m)\Big) = p_{a_1,1} \cdot p_{a_2,2} \cdot \ldots \cdot p_{a_m,m}$$

for any $(a_1, \ldots, a_m) \in \Omega$.

(ii): In order to show the inequality holds, we want to apply Corollary 12.3 from the lecture. To be more specific, the inequality automatically holds when Ω is a distributive lattice, for which $\mathbb{P}$ is log-supermodular, and A_I, A_J are monotonically increasing events. It is known from the previous problem that $\Omega = L_{n,m}$ is a distributive lattice; hence, it remains to show that $\mathbb{P}$ is log-supermodular, and A_I, A_J are monotonically increasing events.

It is known that $\mathbb{P}$ is log-supermodular when $\mathbb{P}(x) \geq 0$ for all $x \in \Omega$, which is always true since $\mathbb{P}$ is a probability measure, and the following inequality holds for $x, y \in \Omega$

$$\mathbb{P}(x)\mathbb{P}(y) \leq \mathbb{P}(x \vee y)\mathbb{P}(x \wedge y) \tag{3.1}$$

Let $x = (a_1, \ldots, a_m)$ and $y = (a_1^{'}, \ldots, a_m^{'})$. Consider the LHS of the equation (3.1), we have

$$\begin{aligned} \mathbb{P}(x)\mathbb{P}(y) &= \mathbb{P}(a_1, \ldots, a_m)\mathbb{P}(a_1^{'}, \ldots, a_m^{'}) \\ &= \prod_{i=1}^{m} p_{a_i,i} p_{a_i^{'},i} \end{aligned} \tag{3.2}$$

It is worth to notice that $a_i \leq a_i^{'}$ or $a_i \geq a_i^{'}$ for $1 \leq i \leq m$. Hence, we can rewrite the equation (3.2) as follows

$$\mathbb{P}(x)\mathbb{P}(y) = \prod_{i=1}^{m} p_{\max(a_i,a_i^{'}),i} p_{\min(a_i,a_i^{'}),i} \tag{3.3}$$

Moreover, by the definition of the operations $\vee$ and $\wedge$ in $L_{n,m}$ in the previous problem, we have

- $\prod_{i}^{m} p_{\max(a_i,a_i^{'}),i} = \mathbb{P}\Big(\big(\max(a_i, a_i^{'})\big)_{i=1,\ldots,m}\Big) = \mathbb{P}(x \vee y)$

- $\prod_{i}^{m} p_{\max(a_i, a_i'), i} = \mathbb{P}\Big(\big(\min(a_i, a_i') \big)_{i=1,\dots,m} \Big) = \mathbb{P}(x \wedge y)$

Hence, $\mathbb{P}(x)\mathbb{P}(y) = \mathbb{P}(x \vee y)\mathbb{P}(x \wedge y)$, and thus the inequality (3.1) is true. Therefore, $\mathbb{P}$ is log-supermodular.

In order to show that A_I is monotonically increasing event, we want show that for $(a_1, \dots, a_m)$ and $(a_1', \dots, a_m')$ in A_I such that $(a_1, \dots, a_m) \leq (a_1', \dots, a_m')$, then $\sum_{i \in I} a_i' \geq \sum_{i \in I} a_i \geq t_I$. This is indeed true, since these tuples are in A_I, then $\sum_{i \in I} a_i' \geq t_I$ and $\sum_{i \in I} a_i \geq t_I$. Moreover, since $(a_1, \dots, a_m) \leq (a_1', \dots, a_m')$, then $a_i \leq a_i'$ for $1 \leq i \leq m$. Therefore, $\sum_{i \in I} a_i' \geq \sum_{i \in I} a_i \geq t_I$, and A_I is monotonically increasing event. Similarly, we can show that A_J is monotonically increasing event.

Therefore, the Corollary 12.3 in the lecture is satisfied, the proof is finished. □